

A digital ride record system for everyday street-hailed pedicabs and tricycles in Tagum City.

Tagum City Focus

Privacy by Design

No Driver App Needed



STREET FLEET



CITY TERMINAL

1 THE PROBLEM

Street-hailed pedicab and tricycle rides usually leave no digital record. Passengers may forget the plate number, MTOP number, ride time, or vehicle details.

This becomes a problem when they leave an item inside, need to identify the vehicle later, or want to report an incident.



2 BENEFICIARIES

Primary:

Students, workers, night commuters, regular pedicab passengers, visitors.



Secondary:

Families, drivers, transport associations, Tagum Traffic Management Office, schools, LGUs.



3 THE SOLUTION

TalaRide lets passengers scan a street-hailed pedicab or tricycle using their phone camera.

The app detects visible vehicle identifiers such as the plate, MTOP, or body number and creates a private digital ride receipt with date, time, and optional location.



4 HOW IT WORKS

1 Scan

Point camera at vehicle.



2 AI Suggest

AI detects plate or MTOP number.



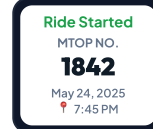
3 Confirm

Passenger verifies or edits details.



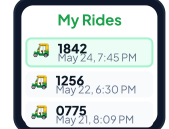
4 Start Ride

Private ride receipt is created.



5 Save

Stored locally on rider's device.



5 COMMUNITY LOST-ITEM RELAY

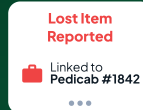
Turning a ride record into a community safety net.

- ✓ No Driver App
- ✓ No QR Required
- ✓ No Booking Required

1 Passenger A scans Pedicab #1842 and later reports a lost item.



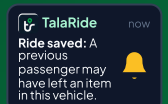
2 An anonymous, expiring matching token is generated.



3 If another TalaRide user later scans the vehicle, they get a discreet relay prompt.



4 A privacy-safe notification connects them with zero location tracking.



6 KEY FEATURES

- Smart Ride Scanner
- Digital Ride Receipt
- Ride History
- Lost Item Mode
- Community Relay
- Trusted Contact
- Location Context
- Privacy First

7 EMERGING TECHNOLOGIES

- Computer Vision
- OCR
- On-device AI (DeepSeek, Gemma)
- Geospatial Technology
- Smart Matching Engine
- Privacy-Preserving Systems

8 TECH STACK

- React Native
- OpenCV
- Expo
- Convex
- TypeScript
- SQLite
- VisionCamera
- MapLibre / OSM
- ML Kit / Lite
- Notifications

CCE 106



PROJECT PROPONENTS

- LEAD Ryan Deniega — Full-Stack & System Lead
- DEV Co-Researcher — Mobile & OCR Integration
- QA Co-Researcher — Field Testing & UX

Adviser: Engr. / Prof. Project Mentor
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REFERENCES & CITATIONS

- [1] City Government of Tagum (2024). *Comprehensive MTOP & Pedicab Regulatory Code*. City Transport Office.
- [2] DeepSeek AI & Google Gemma (2024). *On-Device Lightweight Vision & Reasoning Models for Mobile Platforms*.
- [3] MapLibre / OpenStreetMap (2024). *Offline Vector Tile Mapping for Public Transportation Systems*.
- [4] National Privacy Commission (2023). *RA 10173 Guidelines on Privacy by Design & Data Minimization for Edge Mobile Transit*.

PROJECT GOAL

To turn an ordinary street-hailed ride into a searchable digital ride record using computer vision and mobile technology.

TalaRide — Remember every ride.

Your ride may be informal. Your record doesn't have to be.